# Classification: Predicting Whether the Closing Price of NYSE Composite Index (NYA) will Increase or Decrease Based on Previous Days' Metrics

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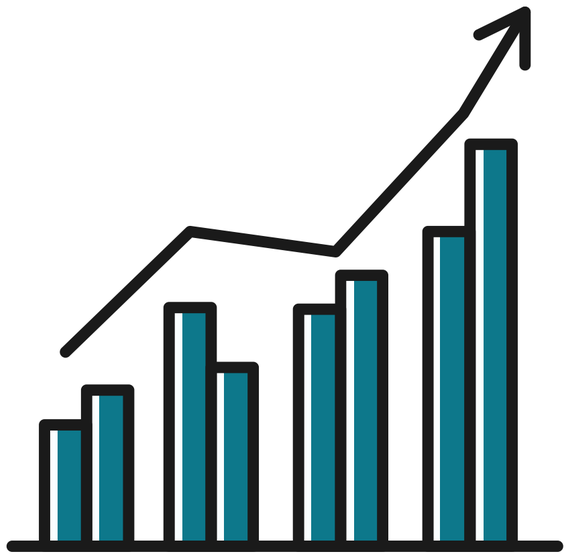
# Business Objective

**Goal:** To develop a model that provides predictions on whether the closing price of the NYA will increase or decrease on the next trading day.



**Why?** If the model performs reasonably well (i.e., a model that is relatively successful in predicting the outcome of increase or decrease the next trading day correctly), there may be a potential use case for traders, such as hedge fund managers, to leverage the model to implement certain trading strategies/decisions for their funds.

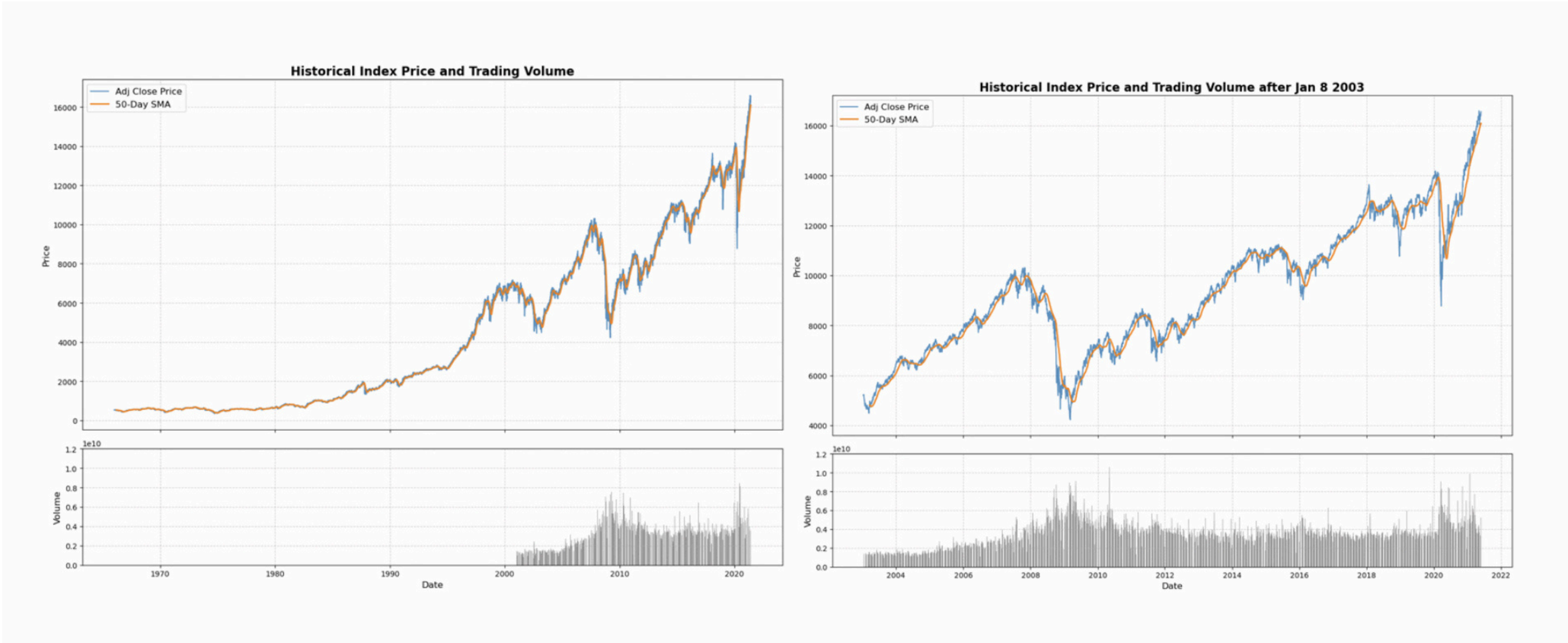
**How?** Given that the model is designed to make daily predictions on stock index movements, it may be more suited for certain daily trading strategies where there is potential to take profit on a daily basis. Future trading, where positions are marked to market daily based on daily price fluctuations, may be a good use case for utilizing a classification model as traders can make buy/sell decisions daily to potentially take profit.

# NYA Index analysis

## Data Overview and Analysis

**The orange SMA_50 line represents a 50-day historical average.**

**Plotted Adj Close price over time**



# Risks and Assumptions

**Risks & Uncertainties** :

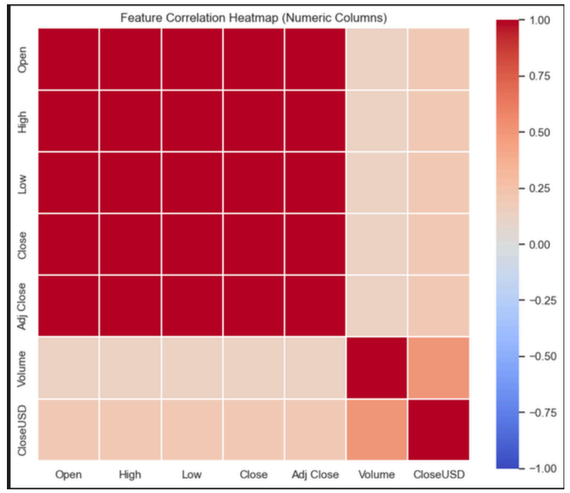
- Dynamic and subject to structural changes
- Massive impact of external factors
- Homoskedasticity (high volatility)

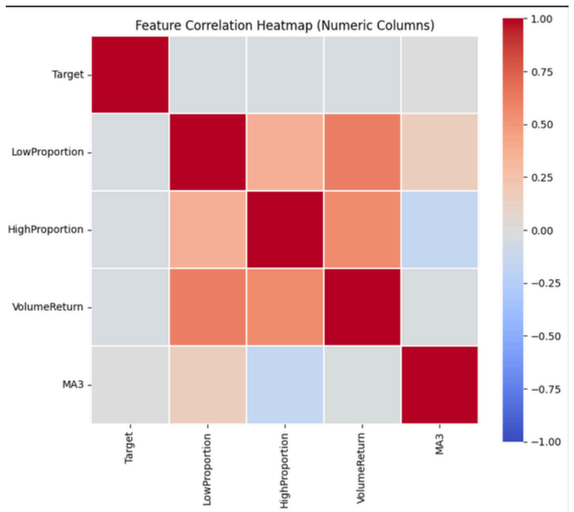
**Assumptions:**

- Model Choice
- 2008 Economic crisis (Uncertainty)
- Data Leakage
- Dataset origin

# Feature Engineer

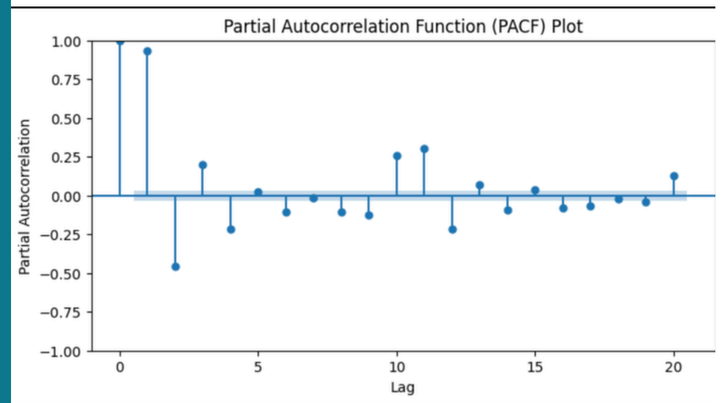
- **Data filtering** - Filtered data based on analysis and set the date as the index.
- **Temporal alignment** - Set a column for tomorrow's close into today's row and created our target column with 2 classes (increase and decrease).
- **Feature creation and section** - After understanding the correlation between features, we created additional features.
- **PACF-based lag selection** - Used an autoregressor to select lags such that the lags are only directly relevant.
- **Train/Test Split** - Performed time series data split such that train and test sets have equal amounts of each class.

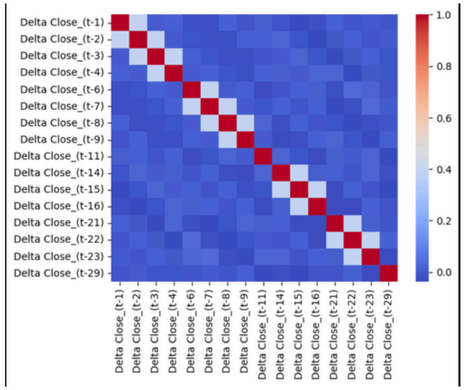



$$LowProportion = \frac{Low}{LastClose}$$

$$HighProportion = \frac{High}{LastClose}$$



# Model Selection and Evaluatuon

- **Cross-Validation -** 5-fold CV with ~20% of the dataset used for validation in each fold.
- **Hyperparameter Tuning**: Used negative log loss (neg_log_loss) to ensure well-calibrated probability predictions.
- **Optimization**: Adam optimizer for gradient descent to achieve stable and efficient convergence.
- **Performance Metrics**: Evaluated models using accuracy and ROC-AUC.
- **Baseline Model**: Random Forest selected as the baseline due to its best overall performance across train and validation sets.

| | Model | Train Accuracy | Validation/Test Accuracy | Train ROC-AUC | Validation/Test ROC-AUC | Notes |
|---|---|---|---|---|---|---|
| 0 | Random Forest | 0.752 | 0.678 | 0.838 | 0.751 | High accuracy and ROC-AUC on train; slight overfitting; best overall performance |
| 1 | Logistic Regression | 0.711 | 0.683 | 0.784 | 0.743 | Moderate train/test metrics; consistent performance; less overfitting than RF |
| 2 | XGBClassifier | 0.757 | 0.682 | 0.837 | 0.744 | Low metrics; overfitting to large extent |
| 3 | LSTM | 0.546 | 0.535 | 0.511 | 0.519 | Low metrics; underfitting; not capturing sequence patterns effectively |

# Conclusion

**Main Takeaways**

- The training and testing results show promise. More explorations can be done to determine whether there are other machine learning models out there that will do a better job of addressing issues with noisy stock market data and non-stationarity
- Simulation of fund performance if the fund were to buy/sell futures contracts on NYA based on the models' predictions is needed to further validate business case, but the initial results seem promising and the strategy may be worthwhile with additional risk oversight measures built in.

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